

A top-down view of a light teal desk. In the top left is a silver laptop with a black keyboard. To its right is a white mouse and a cup of dark coffee. In the bottom left is a dark brown leather notebook with a white pen and a pair of tortoiseshell glasses.

# AIClub





# Modelling

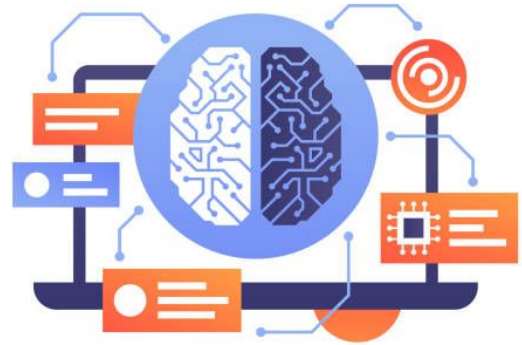
# What is Modelling?

- After exploring and preparing the data, we build a system that can make predictions.
- This stage is called Modelling.
- In this stage, we create an AI model that learns from data.



# What is a Model?

- **A Model is a system that:**
  - Learns from data
  - Identifies patterns
  - Makes predictions or decisions.



## Example:

- Predicting whether a student is “Pass” or “Fail” based on marks.

# Types of Models

- In Class IX AI, we mainly study:
  - Rule-Based Model
  - Learning-Based Model



**Rule-Based Model**

Emphasizes structured approaches

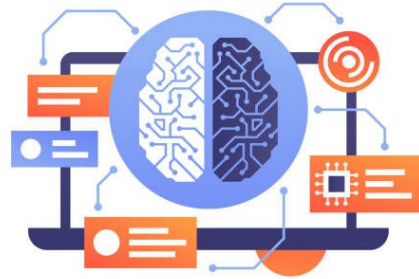


**Learning-Based Model**

Focuses on adaptability and continuous learning

# Rule-Based Model

- A Rule-Based Model works using predefined rules.
  - Human writes the rules
  - Machine follows the rules
  - No learning from data.



Example Rule:

- IF marks > 40 THEN Result = Pass ELSE Result = Fail.

Machine only follows instructions.

# Characteristics of Rule-Based Model

- Simple and easy to design
- Works well for simple problems
- Cannot improve automatically
- Limited flexibility.



Used when rules are clear and fixed.

# Learning-Based Model

A Learning-Based Model learns patterns from data.

- Uses training data
- Improves with experience
- Can handle complex problems.

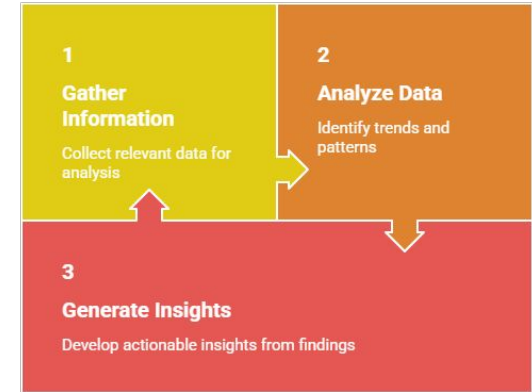
It does not need fixed rules.



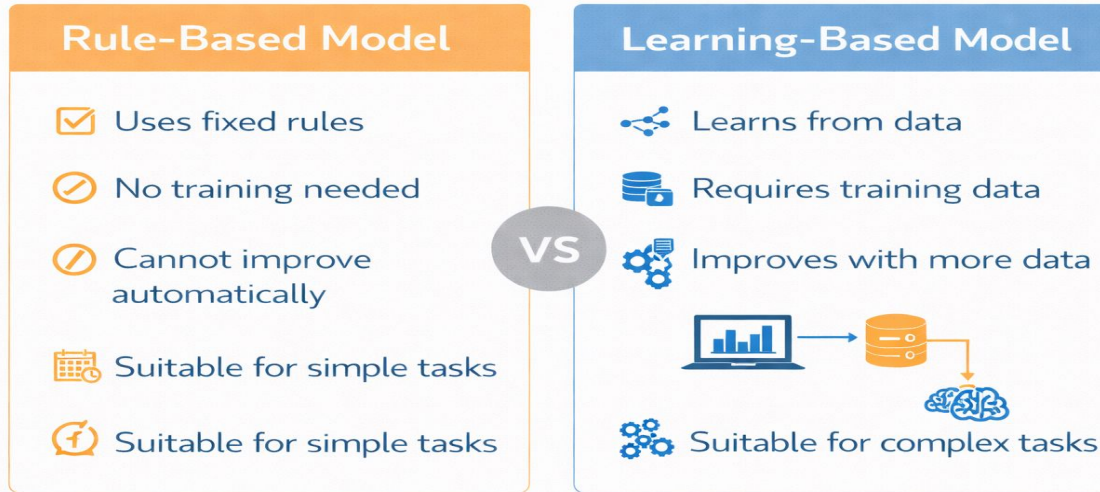
# How Learning-Based Model Works

- Step 1: Provide Training Data
- Step 2: Model learns patterns
- Step 3: Model makes predictions on new data

Example: Instead of writing rules for spam detection, the model learns from thousands of emails.



# Comparison – Rule-Based vs Learning-Based



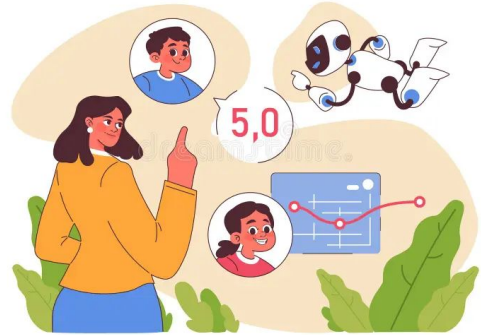
# Example – Student Performance Prediction

Rule-Based Approach:

IF attendance > 75% → Good ELSE → Needs Improvement.

Learning-Based Approach:

- Model learns relationship between:
  - Attendance
  - Study hours
  - Test scores



Then predicts performance.

# Why Modelling is Important?

- ✓ Converts data into intelligence
- ✓ Helps make predictions
- ✓ Solves real-world problems.

Without modelling, AI cannot function.

# Key Takeaways

- ✓ Modelling is Stage 4 of AI Project Cycle.
- ✓ Two main types: Rule-Based and Learning-Based. Rule-Based models
- ✓ follow fixed rules. Learning-Based models learn from data. Learning-Based
- ✓ models are more powerful for complex tasks.

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# THANK YOU



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